

COMPSYS302: Python

1. Introduction

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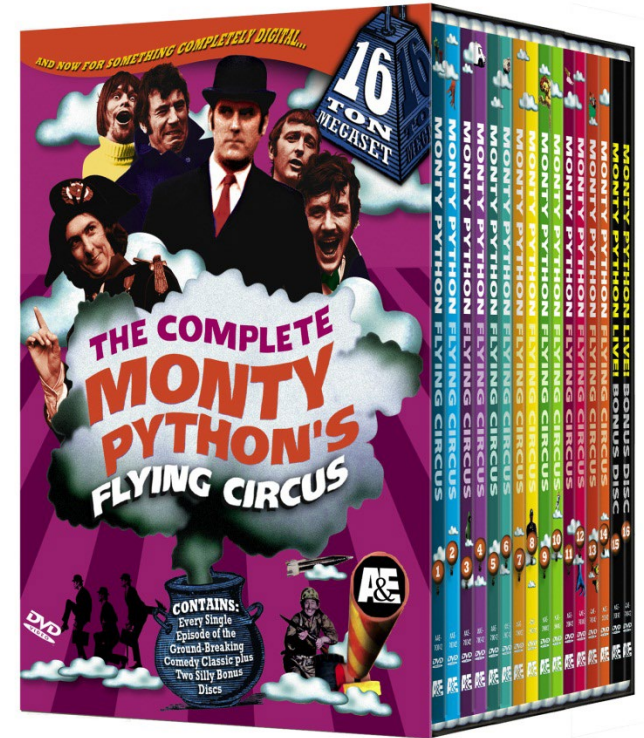
**THE UNIVERSITY
OF AUCKLAND**

FACULTY OF ENGINEERING

1.1 Outline of Python

□ Python is..

- Created by Guido van Rossum in 1991.
 - From "Monty Python's Flying Circus"
 - Interpreter language
 - Icon:  python™
- Use
 - One of 3 development languages of Google.
 - Google, Yahoo, NASA, YouTube, Dropbox, Eve Online, many others
 - The number of users has been on the increase



1.1 Outline of Python

□ Features of Python

- Legibility
 - Simple syntax, easy to read & write code
- Library
 - plentiful library can be used
- Extensibility
 - Extensible in C, C++, C#, Java, ...
- Open Source
 - Managed by Python Software Foundation
 - Python Software Foundation License, which is free to use

1.1 Outline of Python

□ Features of Python

- Dynamic language
 - ➔ At runtime, execute many common programming behaviours that static programming languages perform during compilation
- Slow to run
 - ➔ Not faster than others, but it is quite fast enough
- Rapid to develop
 - ➔ Let programmer focus on higher level design/algorithms rather than getting bogged down in pointer bugs
 - ➔ Very quick for producing a prototype of a new idea

1.1 Outline of Python

□ **Types of Python**

- There are different implementations of Python
- Cpython
 - Written in C
 - The most common Python
- Jython
 - Written in Java
 - Possible to be worked on virtual machine
 - Possible to use graphic libraries, such as Swing, AWT
- IronPython
 - Written in C#
 - Developed for .Net
- pythonCE
 - For WinCE environment

1.1 Outline of Python

□ Version of Python we will use

- We will use version 3 of Cpython, Python 3

□ Difference between Version 2.x and 3.x

- There are some difference
- Example 1) Dividing results
 - Ver. 2.x: **int**

```
>>>3/2  
1
```
 - Ver. 3.x: **float**

```
>>>3/2  
1.5
```
- Example 2) print function
 - Ver. 2.x: **print "a"**
a
 - Ver. 3.x: **print ("a")**
a

1.1 Outline of Python

□ Using Python

- Interactive mode
 - At the command line: `> python`
`>>> 9/2`
`4.5`
- Script mode
 - Write in a file, for example `main.py`
`> python main.py`
- IDE mode
 - For example: PyThon extension for VSCode / PyDev for Eclipse (an Eclipse plugin)
 - Saves in a `"xxx.py"` file, with a console view too
 - You can probably find other IDEs to use

1.1 Outline of Python

□ **How to learn Python**

- Plenty online tutorials and books in library
- Download the Python Documentation and browse offline
 - <http://www.learnpython.org>
 - <http://docs.python.org/download.html>
 - Tutorial
 - Library Reference
 - Language Reference
- Ask TAs in the Labs
 - Weekly Labs, which is not enough, but is very helpful

1.1 Outline of Python

□ Scope of Python in COMPSYS302

- Introduction to Python
- Core Python Language
 - Data types and Operations (variables, list, set, tuple, dictionary)
 - Functions and arguments
 - Module
 - Input and Output includes file I/O
 - Control flow and iterators (if, else, for, while)
 - Exception handling
 - Class
- Python GUI
 - PyQt

1.1 Outline of Python

□ **Python in the classroom**

- Live coding
 - A lot of examples (using VSCode)
 - But, you may feel a bit boring and/or not enough for your project!
- You should spend more hours for practicing by yourselves

□ **Python in the LABs**

- Not compulsory, but helpful if you come
- Practical works and tutorials, such as
environmental setup, question on errors, etc.
- But, TAs may not be able to answer to all questions, especially for the project.